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Unit – 4 : Session – 6 : SLO – 2 | Binary Search Trees — Searching Operation

■ I. Fill in the Blanks

- A Binary Search Tree (BST) is a BINARY TREE with a special property.
- All left descendants must be LESS than the node.
- All right descendants must be GREATER than the node.
- BSTs provide EFFICIENT ($O(\log n)$) searching.
- BSTs are used in DATABASE INDEXING and DICTIONARY (lookup/sorting) applications.
- In a BST, the left subtree contains nodes SMALLER THAN the root node.
- The right subtree contains nodes GREATER THAN the root node.
- Search begins at the ROOT node.
- If the value to be searched is less than root, go to the LEFT subtree.
- If the value to be searched is not found, the result is NULL (NOT FOUND).

■ II. Algorithm to Search a Value in a BST

- SearchElement (TREE, VAL)
- Step 1: IF TREE -> DATA = VAL OR TREE = NULL
- Return TREE
- ELSE
- IF VAL < TREE -> DATA
- Return searchElement(TREE -> LEFT, VAL)
- ELSE
- Return searchElement(TREE -> RIGHT, VAL)
- [END OF IF]
- [END OF IF]
- Step 2: END